Fiscal Year:	FY 2021	**Task Last Updated:**	FY 03/31/2021
PI Name:	Yodh, Arjun Ph.D.		
Project Title:	Crystal-to-Glass Transitions in Colloids (Flight Experiment)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-01
Start Date:	12/19/2018	**End Date:**	01/31/2022
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 1/31/2022 per NSSC information (Ed., 9/20/21) NOTE: End date changed to 12/18/2021 per NSSC information (Ed., 3/12/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	This proposal builds on and continues the research described in Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3) (Ed. note: see in Task Book "Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3)"; grant NNX13AL27G ; Principal Investigator Arjun Yodh). The roots of the work derive from an experimental program that was peer-reviewed and accepted for flight on the International Space Station (ISS); the instrumentation at the heart of the original flight proposal for ISS was part of the so-called Light Microscopy Module (LMM). More recently, a Science Concept Review (SCR) for Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 2) was held in June 2011, and the proposed research was strongly endorsed by the External Review Panel in their report on June 14 2011; the science goals and experimental plans for Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3), were also critically reviewed. The current research will be integrated into the hardware developed for the spaceflights that are named by variants of the term “ACE” (e.g., ACE-H1). In addition, related complex fluid measurements have been carried out in the Binary Colloid Alloy Test (BCAT) BCAT-series flight experiments. In response to these developments, the science aims have evolved to take advantage of important recent breakthroughs in the field. Herein we describe these complex fluid experiments. Note also, whenever possible, we will be joining forces with our colleagues from Europe to perform collaborative experiments within the European Space Agency (ESA) and NASA facilities. The overall rationale for these investigations is that the elimination of particle sedimentation effects in microgravity creates a purely “thermodynamic” environment for colloidal suspensions wherein particle size, shape, and volume fraction, and interparticle interactions are the dominant determinants of the assembled structures. This feature enables us to make new materials that do not form on Earth and to elucidate the effects of gravity in soft matter science that are masked in ground-based experiments. This version of Low Volume Fraction Entropically Driven Colloidal Assembly explores a range of experimental complex fluids with themes related to and evolved from previous proposal aims. The original studies were centered around entropically driven low volume-fraction self-assembling complex fluids, suspensions composed of an ensemble of large particles, and additional smaller constituents (either a small colloidal particle, a polymer, or a micelle). Phase 3 studies retained many of the original themes but also had new focus that reflected the evolution of the complex fluid field, as well as new developments in our laboratory. Broadly, the proposed research will elucidate a wide-range of phenomena in condensed matter physics that has implications for phase-diagrams of entropically driven colloidal media, crystallization, and melting mechanisms in three-dimensional (3D) bulk media and in finite volumes, jamming, and glass formation in 3D, and, opportunistically, crystallization and melting mechanisms and aging of colloids/soft-matter based on more anisotropic particles and filaments on a variety of length scales. International collaborators are Prof. Luca Cipelletti, Université Montpellier II, Montpellier, France; Prof. Roberto Piazza, Department of Nuclear Engineering, Politecnico di Milano, Milan, Italy; Prof. Willem Kegel, Van 't Hoff Laboratory for Physical and Colloid Chemistry, Debye Research Institute, Utrecht University, Utrecht, The Netherlands.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research can provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.
Task Progress:	The research involves ground-based complex fluid experiments, as well as colloidal gel experiments we completed on the International Space Station (ISS) during this time period. The highlight of our accomplishments was the ISS experiment in which six colloidal particle suspension samples were loaded and investigated on two modules (ACE-T4-1 and ACE-T4-2). From April 20, 2020 through September 4, 2020, approximately six terabytes of science data were collected. This data was largely derived from microscope images of the samples as a function of particle packing fraction, colloid composition, and temperature. The rationale for microgravity stems from the elimination of particle sedimentation effects. Elimination of sedimentation creates a purely “thermodynamic” environment for colloidal suspensions wherein particle size, shape, volume fraction, and inter-particle interactions are the dominant determinants of the assembled structures and their dynamics. This attribute enables us to elucidate the effects of gravity that are masked in ground-based experiments; successful elimination of the effect of gravity, in turn, makes the findings of model colloid experiments (such as ours) more relevant to the behavior of atomic/molecular matter wherein gravity is less important (because atomic and molecular masses are tiny). The lack of gravitational stresses can also help us identify new pathways to make soft materials that do not readily form on Earth. In the case of colloidal gels and attractive glasses that we have studied on ISS, the microgravity experiments permit confocal microscopy imaging of the colloids and thereby offer a window into the structure and dynamics of gels (or attractive glasses) at high packing fraction, without the complications of gravitational stresses that arise on Earth. In total, the insights gained could be important for synthesis and control of gel-based materials that range from everyday commercial products such as paints and toothpaste, to complex fluids for 3D printing (including 3D printing in space). Briefly, the experiments performed on ISS investigate the character, microstructure, and mechanical response of dense colloidal gels in microgravity (note, such materials might also be considered to be attractive colloidal glasses). The samples studied on ISS were modified somewhat with respect to the repulsive glasses initially prepared on the ground; as a result, we refocused the ISS research to opportunistically probe questions about colloidal gels and attractive glasses, i.e., important questions about gels and attractive glasses that differed somewhat from our original questions about repulsive glasses. Nevertheless, the systems studied still fall into the category of disordered soft solids, and we still sought to understand their structural and dynamical characteristics. The six samples had volume fractions of ~0.58 at ambient temperature. The colloidal samples consisted of two particle types, a dominant particle (occupying most of the sample volume) and a dopant particle (which introduced different degrees of disorder into the samples). The microgravity environment eliminated directional gravitational stresses that would otherwise naturally arise on Earth in colloid experiments at high packing fraction. Importantly, our samples were comprised of thermo-sensitive, soft particles (e.g., pNIPAM or NIPA particles). The diameter of these spheres decreases with increasing temperature. This effect enabled us to vary packing fraction and internal stress within the colloidal gels via changes in temperature. Of the six samples, the first had only the primary component, spheres with diameter of ~1.3 micron at ambient temperature; samples #2 to #6 also contained dopants (spheres with ~1.0 micron diameter at ambient temperature) with increasing concentration from 1% to 5%.

	We have begun to convert these six terabytes of collected data into a form that can be processed for image analysis. In a small fraction of the samples we analyzed resulting processed images. We observed general trends with increasing temperature (decreasing packing fraction, even though the number density remains constant). In particular, the gels became more porous and exhibited larger motional fluctuations as the packing fraction decreased. We reconstructed three-dimensional (3D) images of particle positions within a few colloidal gels, and we determined the sample pair correlation functions or structure factors) of these samples. We hope to discern the modes of relaxation of the gels from the dynamical images, and then correlate these factors with the gel's responses to mechanical stress and the gel's packing fraction and degree of disorder.
Bibliography Type:	Description: (Last Updated: 06/16/2025)
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